Problem Set 1

Due: 27 May 2024, 3:30 p.m.

Problem 1. Estimate the number of piano tuners in Beijing. List all steps of your estimation procedure.

(*3 points*)

Problem 2. The formula for the period T of an object, orbiting close to the surface of a planet, is of the form $T = k \cdot \varrho^A \cdot G^B$, where k is a dimensionless constant, ϱ is the planet's density of mass, and G is the gravitational constant. Using dimensional analysis, find the exponents A and B.

(*3 points*)

Problem 3. *True or false?* If the magnitude of vector $\mathbf{u}$ is less than the magnitude of vector $\mathbf{w}$, then the (Cartesian) x component of $\mathbf{u}$ is less than the x component of $\mathbf{w}$.

Explain your answer briefly.

(*2 point*)

Problem 4. Positions of two particles at an instant of time are given by the vectors $\mathbf{r}_1 = 4\hat{n}_x + 3\hat{n}_y + 8\hat{n}_z$ and $\mathbf{r}_2 = 2\hat{n}_x + 10\hat{n}_y + 5\hat{n}_z$, respectively (in the units of m). Find

(a) the magnitudes of these vectors,

(b) the displacement vector $\mathbf{r}_{12}$ from the position of particle 1 to the position of particle 2 and the unit vector in the direction of the vector $\mathbf{r}_{12}$.

(c) the angles between all pairs of vectors $\mathbf{r}_1$, $\mathbf{r}_2$, and $\mathbf{r}_{12}$ (give your answer both in radians and degrees),

(d) the orthogonal projection (vector) of the vector $\mathbf{r}_2$ onto the vector $\mathbf{r}_1$,

(e) the cross product $\mathbf{r}_{12} \times \mathbf{r}_1$.

(*1/2 + 1/2 + 1 + 1 + 1 points*)

Problem 5. A vector $\mathbf{n} = \mathbf{n}(t)$ has unit length for any instant of time t.

(a) Is the magnitude of $\dot{\mathbf{n}}$ equal to one, too? If so, prove it, otherwise find a counterexample.

(b) Show that, for any t, the vectors $\mathbf{n}$ and $\dot{\mathbf{n}}$ are perpendicular to each other.

(*2 + 2 points*)

Problem 6. A particle moves along the x axis with velocity $v_x(t) = v_0 \exp(-t/\alpha)$, where $v_0 = 10$ m/s and $\alpha = 3$ s. At the initial instant $t = 0$ s, the particle is at $x = 0$ m.

(a) Find the acceleration a_x of the particle at any instant of time t.

(b) Find the time after which the particle stops.

(c) What is the distance traveled by the particle from $t = 0$ s until it stops?

(d) Sketch the graphs $a_x(t)$, $v_x(t)$, and $x(t)$.

(e) Find the particle's average velocity between $t = 0$ and $t = 10$ s.

(f) What is its average speed in the same time interval?

(*1/2 + 1/2 + 1 + 3/2 + 1/2 + 1 points*)

Problem 7. A tourists kayaks from marina A down the river to marina B in 3 hours, and from B to A in 6 hours, paddling always with the same speed with respect to the river. How much time does he need to get from A to B without paddling?

(*3 points*)

Problem 8. A spare paddle drops from a fisherman's canoe traveling up the river exactly at the moment as it passes under a bridge. After one hour of paddling the fisherman realizes that the paddle is missing He turns around and paddles his canoe back to find the paddle 6 km down the bridge. Find the speed of the river's current.

Assume that the fisherman paddles always with the same speed with respect to the river. Clearly indicate the frame of reference the problem is being solved in.

(*3 points*)

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problem 1. step 1. the estimated population of Beijing is 21 million in 2024.

step 2. suppose 3 people in Beijing can constitute a family

step 3. Then the number of household is estimated to be

$2100000 \div 3 = 700000$

step 4. according to a report in 2013, per 100 household will there be 4.38 piano

We estimate that in 2024 the percentage will be $5/100$.

then the number of piano in 2024 is $7 \times 10^5 \times 5\% = 35000$

step 5. Estimate that a piano needs to be tuned per year.

step 6. Each year the number of pianos which need tuning is 35000

step 7. a piano tuner may tune about 2 pianos per day, working around 250 days a year

then for each tuner $2 \times 250 = 500$ pianos/year.

step 8. the number of tuner

$35000 \div 500 = 70$.

problem 2. for T the SI unit is s

for ρ: kg/m^3 G: $m^3/(kg\,s^2)$

so $s = (kg/m^3)^A \, (m^3/(kg \cdot s^2))^B$

$$\begin{cases} A = B \\ 2B = -1 \end{cases} \Rightarrow \begin{matrix} B = -\frac{1}{2} \\ A = -\frac{1}{2} \end{matrix}$$

problem 3. False.

Disprove by counter example:

$\vec{u} = (\sqrt{2}, \sqrt{2})$ $|\vec{u}| = 2$ $\vec{w} = (1, 2\sqrt{2})$ $|\vec{w}| = 3 > |\vec{u}|$

But $u_x > w_x$

Both the direction and magnitude are influencing the x component. But the statement considers only the magnitude.

problem 4.

(a) $|r_1| = \sqrt{4^2+3^2+8^2} \approx 9.43\ m$ $\quad |r_2| = \sqrt{2^2+10^2+5^2} \approx 11.36\ m$

(b) $\vec{r}_{12} = \vec{r}_2 - \vec{r}_1 = -2\hat{n}_x + 7\hat{n}_y - 3\hat{n}_z$

the unit vector $\frac{\vec{r}_{12}}{|\vec{r}_{12}|} = \frac{-2\hat{n}_x + 7\hat{n}_y - 3\hat{n}_z}{7.87} = -0.26\hat{n}_x + 0.90\hat{n}_y - 0.38\hat{n}_z$

(c) $\cos\langle \vec{r}_1, \vec{r}_2 \rangle = \frac{\vec{r}_1 \circ \vec{r}_2}{|\vec{r}_1||\vec{r}_2|} = \frac{8+30+40}{9.43 \times 11.36} = 0.728$

$\angle\langle \vec{r}_1, \vec{r}_2 \rangle = \arccos(\cos\langle \vec{r}_1, \vec{r}_2 \rangle) = 43.28° \approx \frac{6}{25}\pi$

$\cos\langle \vec{r}_1, \vec{r}_{12} \rangle = \frac{\vec{r}_1 \circ \vec{r}_{12}}{|\vec{r}_1||\vec{r}_{12}|} = \frac{-8+21-24}{9.43 \times 7.87} = -0.15$

$\angle\langle \vec{r}_1, \vec{r}_{12} \rangle = 98.63° = \frac{11}{20}\pi$

$\cos\langle \vec{r}_2, \vec{r}_{12} \rangle = \frac{-4+70-15}{11.36 \times 7.87} = 0.57$

$\angle(\vec{r}_2, \vec{r}_{12}) = 55.25° = 0.307\pi$

(d) $|\vec{r}_2|\cos\langle \vec{r}_1, \vec{r}_2 \rangle = \frac{\vec{r}_1 \circ \vec{r}_2}{|\vec{r}_1|} = \frac{8+30+40}{9.43} = 8.27\ m$

(e) $\vec{r}_{12} \times \vec{r}_1$

$$\begin{array}{ccc} \hat{n}_x & \hat{n}_y & \hat{n}_z \\ -2 & 7 & -3 \\ 4 & 3 & 8 \end{array}$$

$\vec{r}_{12} \times \vec{r}_1 = (56+9)\hat{n}_x - (-16+12)\hat{n}_y + (-6-28)\hat{n}_z$

$= 65\hat{n}_x + 4\hat{n}_y - 34\hat{n}_z$

problem 5.

(a) No.

counterexample $\quad \vec{n}(t) = \cos(2t)\hat{n}_x + \sin(2t)\hat{n}_y$

$\dot{n}(t) = 2\sin(2t)\hat{n}_x + 2\cos(2t)\hat{n}_y$

$|\dot{n}(t)| = 2\sqrt{2}$

(b) $|\vec{n}(t)| = 1 \qquad \vec{n}(t) \circ \vec{n}(t) = 1$

$\frac{d}{dt}(\vec{n}(t) \circ \vec{n}(t)) = 2\dot{n}(t)\,\vec{n}(t) = 0 \qquad$ so $\dot{n}(t) \cdot \vec{n}(t) = 0$

$\vec{n} \perp \dot{n}$.

problem 6.

(a) $a_x = \dot{V}_x(t) = \frac{d}{dt}\left(V_0 e^{-\frac{t}{\alpha}}\right) = \frac{d}{dt}\left(10e^{-\frac{t}{3}}\right)$

$= 10e^{-\frac{1}{3}t} \times (-\frac{1}{3})$

$= -\frac{10}{3}e^{-\frac{1}{3}t}$ m/s²

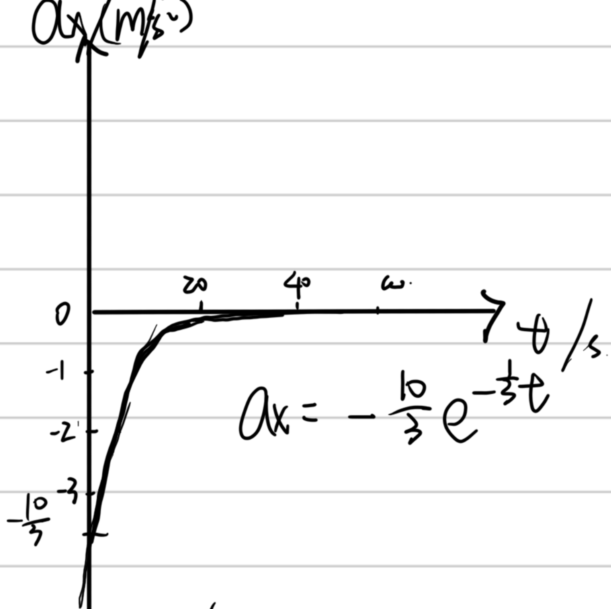
(b) $V_x(t)=0 \Rightarrow 10e^{-\frac{t}{3}}=0$

$t \to \infty$ $\quad V_x(t) \to 0$

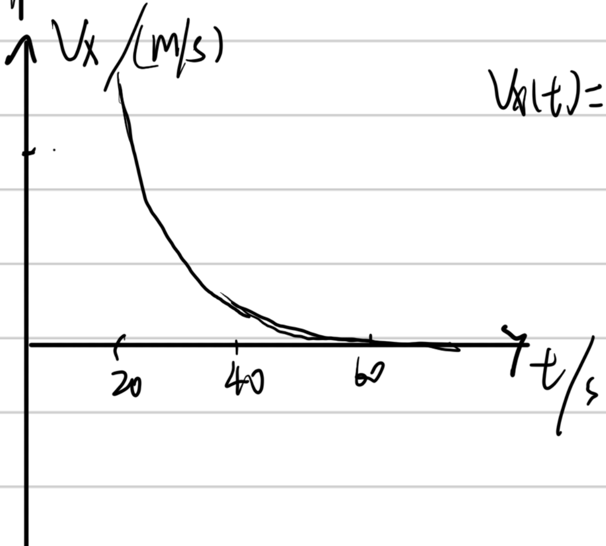
(c) $s = \int_0^\infty |V_x(t)|\,dt = \int_0^\infty 10e^{-\frac{t}{3}}\,dt = -30e^{-\frac{t}{3}}\Big|_0^\infty$

$= 30$ m

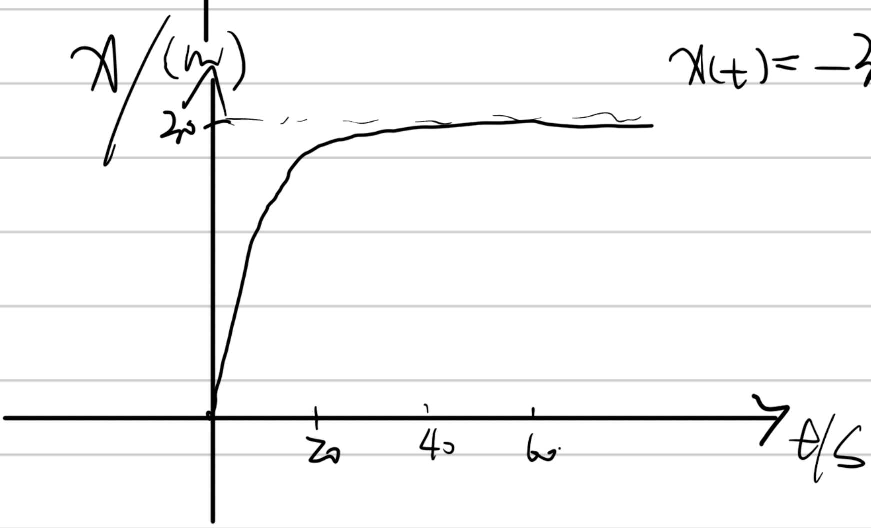
(d) $a_x(t)$



$V_x(t)$ $\qquad V_x(t) = 10e^{-\frac{t}{3}}$



$x(t)$ $\qquad x(t) = -30e^{-\frac{t}{3}} + 30$



(e) $x(t=10s) = -30e^{-\frac{10}{3}} + 30$

$$\bar{v} = \frac{x(t=10s) - x(t=0s)}{t} = \frac{30 - 30e^{-\frac{10}{3}}}{10} = 3 - 3e^{-\frac{10}{3}} \, m/s$$

(f) since the motion of the particle has only one direction

average speed = average velocity $= 3 - 3e^{-\frac{10}{3}}$ m/s

problem 7.

$A \to B$ $(V_p + V_0) t_1 = S$

$B \to A$ $(V_p - V_0) t_2 = S$

$V_p + V_0 = 2(V_p - V_0)$ $\quad V_p = 3V_0 \quad S = 12V_0$

$$t_3 = \frac{S}{V_0} = 12 \, h$$

problem 8.

first.

Take the ground as the frame of the reference, from the point the paddle drops to the point it is raised:

$T = \frac{L}{V_c}$ $\quad L = 6$ km.

from the drop to when the man realize.

$L_{up} = (V - V_c) \cdot t_1$

then. $(T - t_1)(V + V_c) = L_{up} + L$

$V_c = 3$ km/s